Presentation and challenge session

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Presentation and challenge session

Show the aggregate result, failure evidence and honest unknowns rather than only a best demonstration.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of presentation and **challenge session**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Final project manuscript. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of final project manuscript

Reference example, not the only acceptable wording: Teacher explanation: A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run.

Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of peer review and response

Reference example, not the only acceptable wording: Teacher explanation: Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

For each comment record concern, response, change location and new **evidence** if needed. If no new **evidence** exists, narrow the claim instead of promising an unperformed test.

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of figures that preserve **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison.

A plot is a representation of **evidence**, not extra **evidence**. Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot.

Show condition breakdowns and unassessed counts where relevant.

Pictured end goal

Explain presentation and **challenge session** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Final project manuscript

Peer review and response

Figures that preserve evidence

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A presentation should make the **evidence** easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of presentation and **challenge session**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

A presentation should make the **evidence** easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating **evidence** or stating it is missing.

Presentation item	Complete supplied exercise contract
Main study	Fourteen passes among twenty cases
Main displayed item	One perfect output selected for demonstration
Supported main rate	14/20 = 70% under the supplied study summary
Other main cases	Six are not documented as passes; actual failure/unassessed breakdown not supplied here
Separate picture	Camera A only: ten correct of twelve photos
Original camera-B evidence	No records
Changed picture	One camera-B success displayed; selection/coverage not established

Presentation item	Complete supplied exercise contract
Independent challenge	Mobile performance asked about a desktop-only fictional project

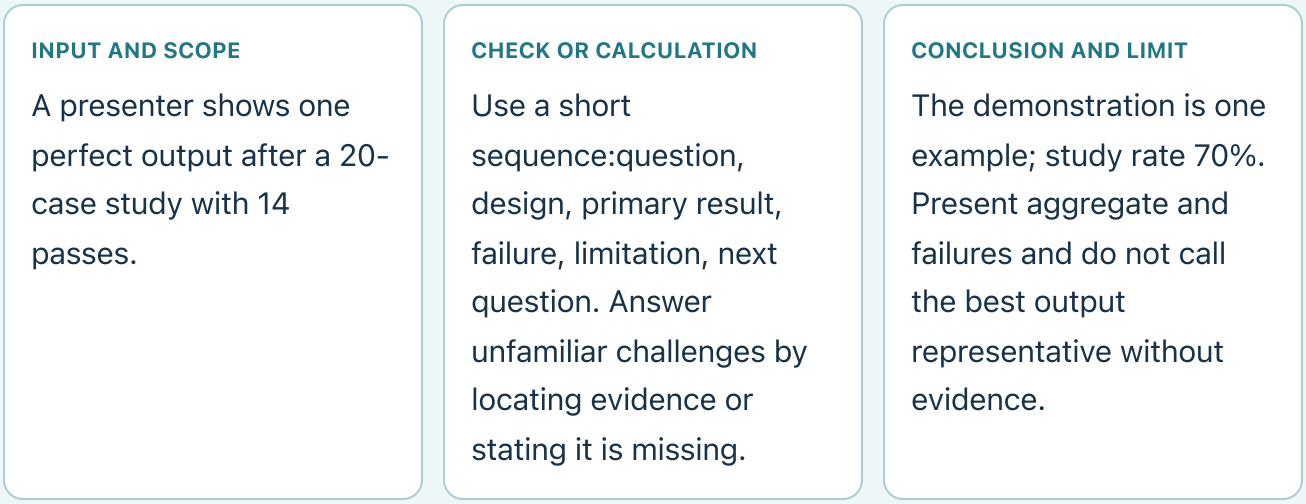
Main evidence ledger and checked key

Main presentation step	Evidence-grounded version
Selected perfect output	Identify as one demonstration
Aggregate primary result	Fourteen of twenty, or seventy percent
Remaining six cases	Show actual failure/unassessed record when available; do not invent breakdown
Claim best output is representative	Not established by selecting it
Failure/limitation slide	Retain concrete supplied evidence and gaps
Next question	Propose a bounded test, with not-run status
Audience question outside scope	Locate evidence or state it is absent

Main selected perfect output is one demonstration; aggregate rate 14/20 = 70%. Six cases are not documented as passes in this summary; do not invent the exact failure/unassessed breakdown. Retain actual negative-case records when supplied. Picture original camera A ten/twelve ≈ 83.333%, no B records. Later one B success is a demonstration with unknown coverage, not broad reliability.

Propose reserved B cases with labels/grouping and fixed scoring, not a fabricated result. Independent fictional desktop-only project cannot answer mobile performance; say it was not tested and specify an appropriate reserved mobile follow-up with fixed criteria.

Presentation and challenge session — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**. Qualification: A confident response should not replace missing **evidence** with speculation. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A presenter shows one perfect **output** after a 20-case study with 14 passes.

The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Presentation audit	Checked interpretation
One perfect output	Demonstration, not the whole study
14/20 study passes	Seventy percent, not perfect performance
Six not documented as passes	Do not invent their exact missing/failure breakdown
No camera-B records originally	State scope; do not promise performance
One later camera-B success	Record separately, not a broad reliability estimate
Question about mobile after desktop-only test	Unknown in that fictional project
Future test proposed	Keep status not run
Honest limitation stated	Retain the primary result; do not hide it behind a demonstration

Separate pictured facts and full reasoning

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

The project tested twelve photos from camera A only. Ten were correct; no camera B records exist.

The separate picture’s original project tests twelve camera-A photos and gets ten correct, approximately 83.333%. It contains no camera-B records, so the original study cannot establish performance on camera B. A proposed follow-up uses reserved camera-B cases with labels, source grouping and a fixed scoring rule. In the changed scenario, one camera-B success is displayed.

That provides a single demonstration, not coverage or a broad reliability estimate; do not erase it or pretend it was part of the original study. Keep its selection status separate and continue to state what a planned representative condition test would need.

Separate challenge evidence	Honest answer
Original camera A	10/12 correct ≈ 83.333% on those photos
Original camera B	No records; no supported rate
Later displayed B success	One demonstrated observation, selection/coverage unknown
Can claim broad camera-B reliability?	No
Proposed test	Reserved labelled B cases, grouping and fixed scoring
Proposed test status	Not run; no result invented
Remaining design issues	Sampling and uncertainty still need justification

Answer the challenge from the evidence you have

Camera A: tested

10 / 12 correct

≈83.333% on supplied photos

Camera B: original study

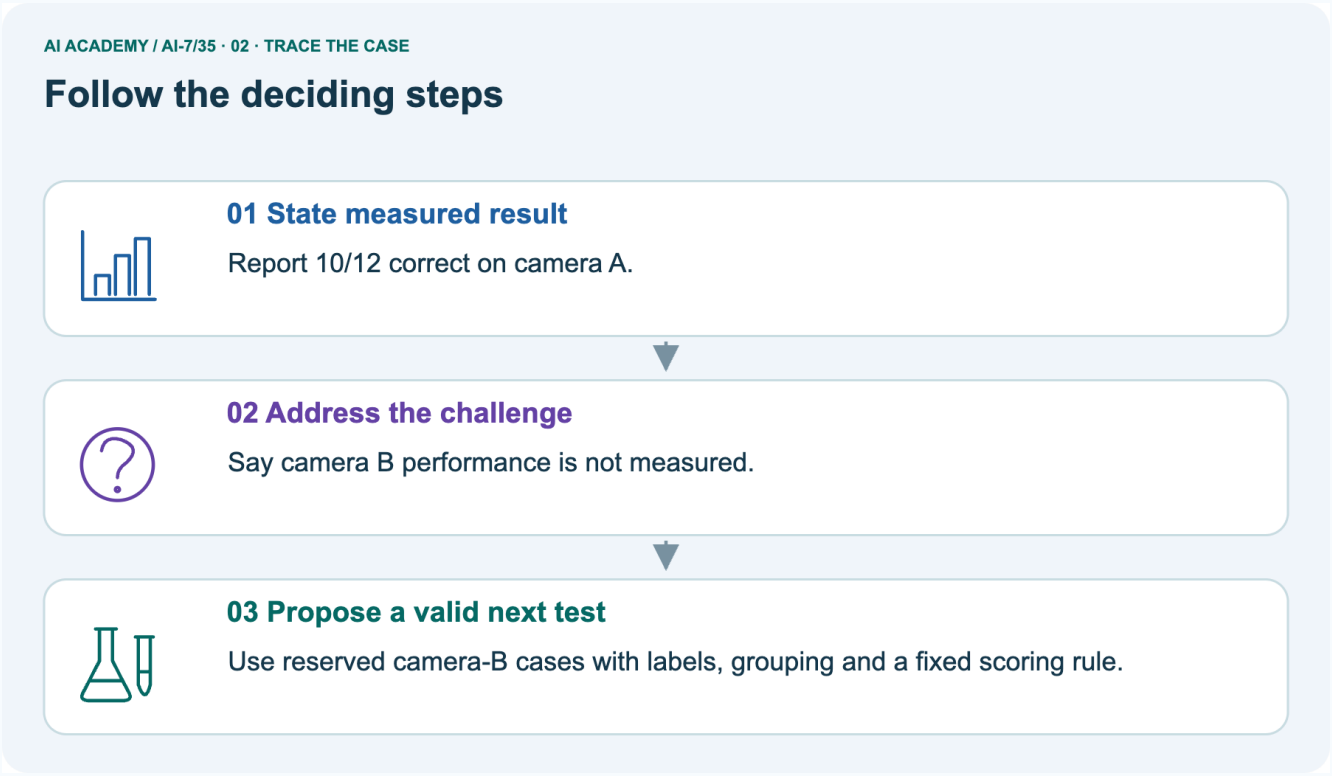
NO RECORDS

Do not promise an untested rate

Later one camera-B success: one demonstration, not broad reliability.

Propose reserved camera-B cases, fixed scoring and recorded grouping.
Mark the proposed test as not run; keep original and demonstration records separate.

Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State measured result	Report 10/12 correct on camera A.
2. Address the challenge	Say camera B performance is not measured.
3. Propose a valid next test	Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

The original result and proposed study still need sampling and uncertainty discussion.

Change: The reviewer shows one camera-B success.

Prediction: That is one demonstration, not an estimate of broad camera-B reliability.

Reason: One selected **observation** does not represent the full proposed condition.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Challenge session	Questions testing the basis of a presentation
Representative example	An example whose selection is justified for the stated purpose
Evidence response	An answer grounded in records or explicit absence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Presentation and challenge session

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A presenter shows one perfect **output** after a 20-case study with 14 passes. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Presentation and **challenge session**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A confident response should not replace missing **evidence** with speculation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

Report 10/12 correct on camera A. Say camera B performance is not measured. Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Changed case: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

Changed-case reasoning: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main selected perfect output is one demonstration; aggregate rate 14/20 = 70%. Six cases are not documented as passes in this summary; do not invent the exact failure/unassessed breakdown. Retain actual negative-case records when supplied. Picture original camera A ten/twelve ≈ 83.333%, no B records. Later one B success is a demonstration with unknown coverage, not broad reliability.

Propose reserved B cases with labels/grouping and fixed scoring, not a fabricated result. Independent fictional desktop-only project cannot answer mobile performance; say it was not tested and specify an appropriate reserved mobile follow-up with fixed criteria.

Diagnose the current misconceptions

The worked result establishes every future case: A confident response should not replace missing **evidence** with speculation.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

Teacher explanation: Mobile behaviour is untested. State desktop scope and propose a new appropriately designed evaluation; do not improvise a positive result.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A presentation should make the **evidence** easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating **evidence** or stating it is missing.

W2 Trace the worked case

A presenter shows one perfect **output** after a 20-case study with 14 passes. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

W3 Audit the limit

A report omits this qualification: A confident response should not replace missing **evidence** with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A confident response should not replace missing **evidence** with speculation.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A confident response should not replace missing **evidence** with speculation.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ **Earlier wording, alternatives and picture transcripts**

Guided and exit questions

Guided: The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**. Qualification: A confident response should not replace missing **evidence** with speculation. Exit: accept a concrete

variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

Teacher explanation: Mobile behaviour is untested. State desktop scope and propose a new appropriately designed evaluation; do not improvise a positive result.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Mobile behaviour is untested. State desktop scope and propose a new appropriately designed evaluation; do not improvise a positive result. Require intermediate reasoning and the stated scope.

Fresh retry: Audience asks for an unmeasured energy comparison. Expected reasoning: Say energy is unmeasured; explain what resource instrumentation and matched comparison would be needed.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Research defense and next questions. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of presentation and **challenge session**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: A presenter shows one perfect **output** after a 20-case study with 14 passes. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Check every learner before discussion: Use the worked case for Presentation and **challenge session**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A confident response should not replace missing **evidence** with speculation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The reviewer shows one camera-B success. Require a **prediction** and a reason before comparing. Expected result: That is one demonstration, not an estimate of broad camera-B reliability.

Reason: One selected **observation** does not represent the full proposed condition.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Presentation and challenge session: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

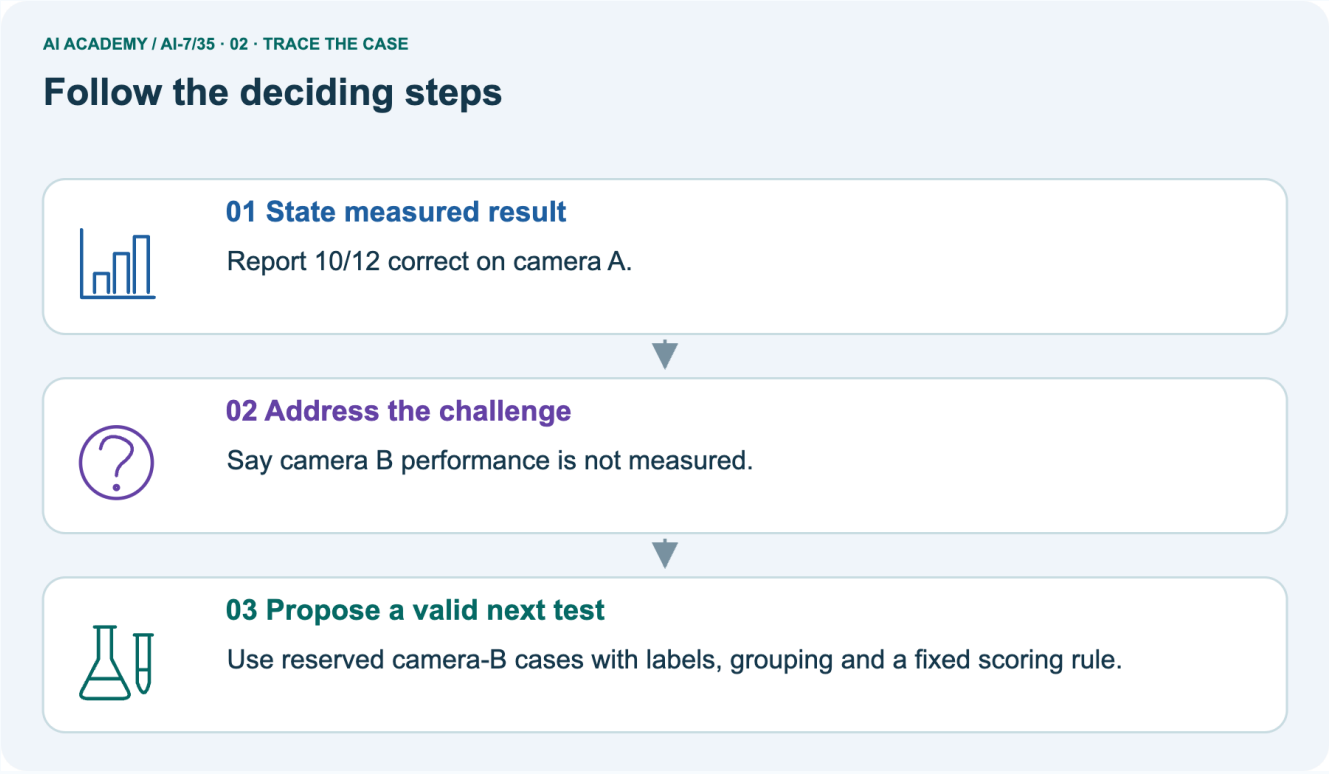
Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera. Supplied facts: The project tested twelve photos from camera A only.

Ten were correct; no camera B records exist.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



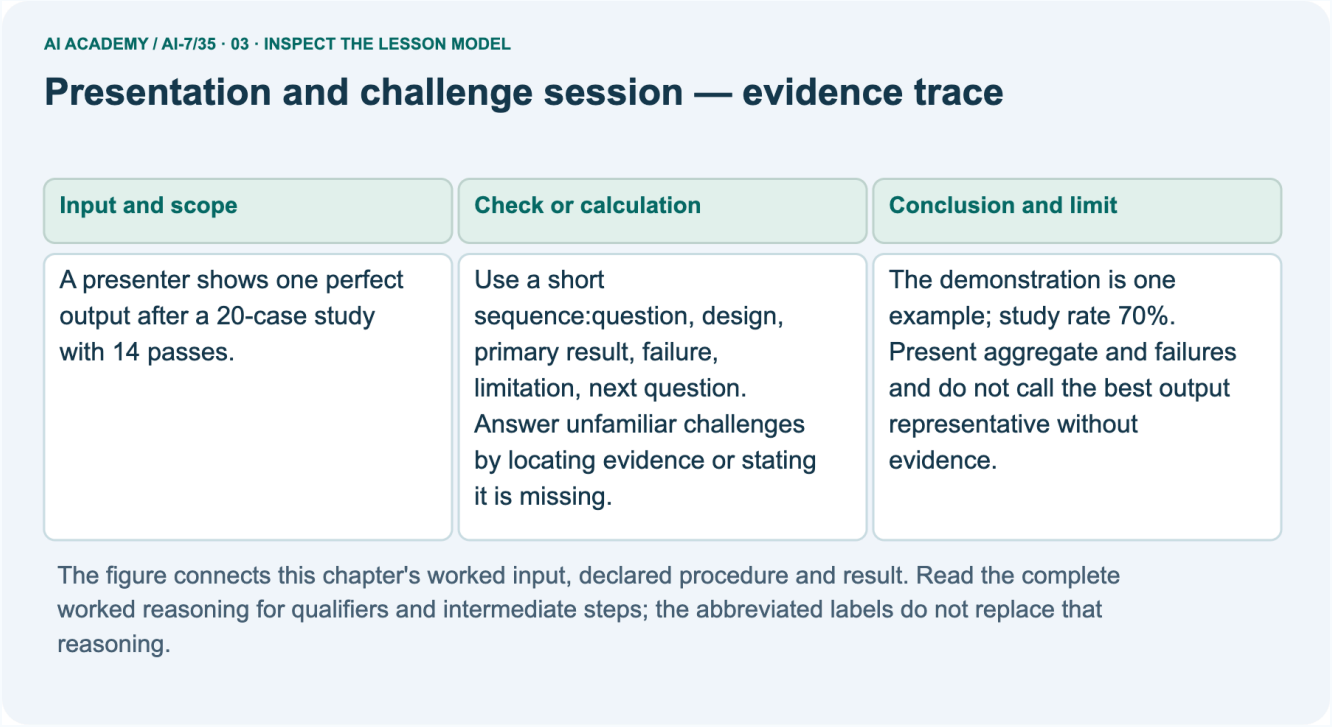
Follow the labelled picture with the matching explanation. Open the vector diagram

State measured result: Report 10/12 correct on camera A. → Address the challenge: Say camera B performance is not measured. → Propose a valid next test: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Presentation and challenge session — evidence trace



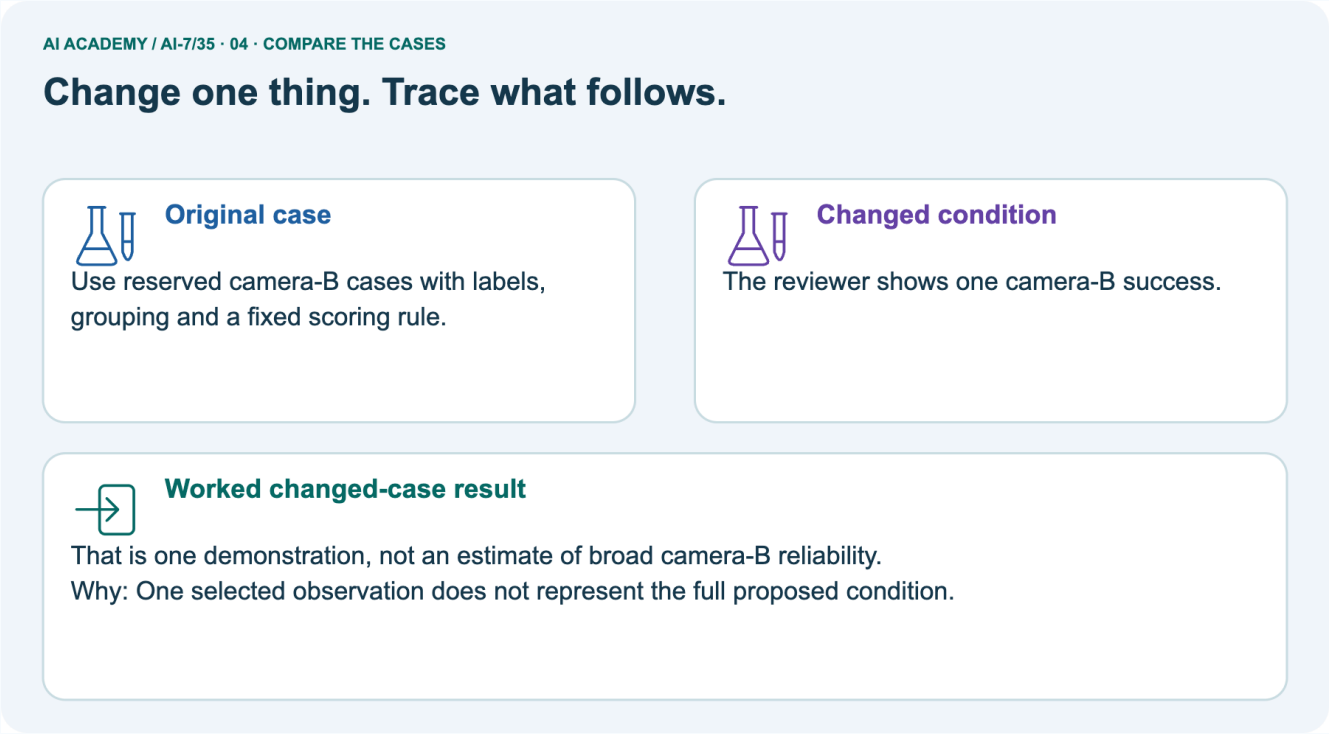
Follow the labelled picture with the matching explanation. Open the vector diagram

Presentation and **challenge session** — **evidence** trace. A presenter shows one perfect **output** after a 20-case study with 14 passes. | Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating **evidence** or stating it is missing. | The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.
Changed condition: The reviewer shows one camera-B success.

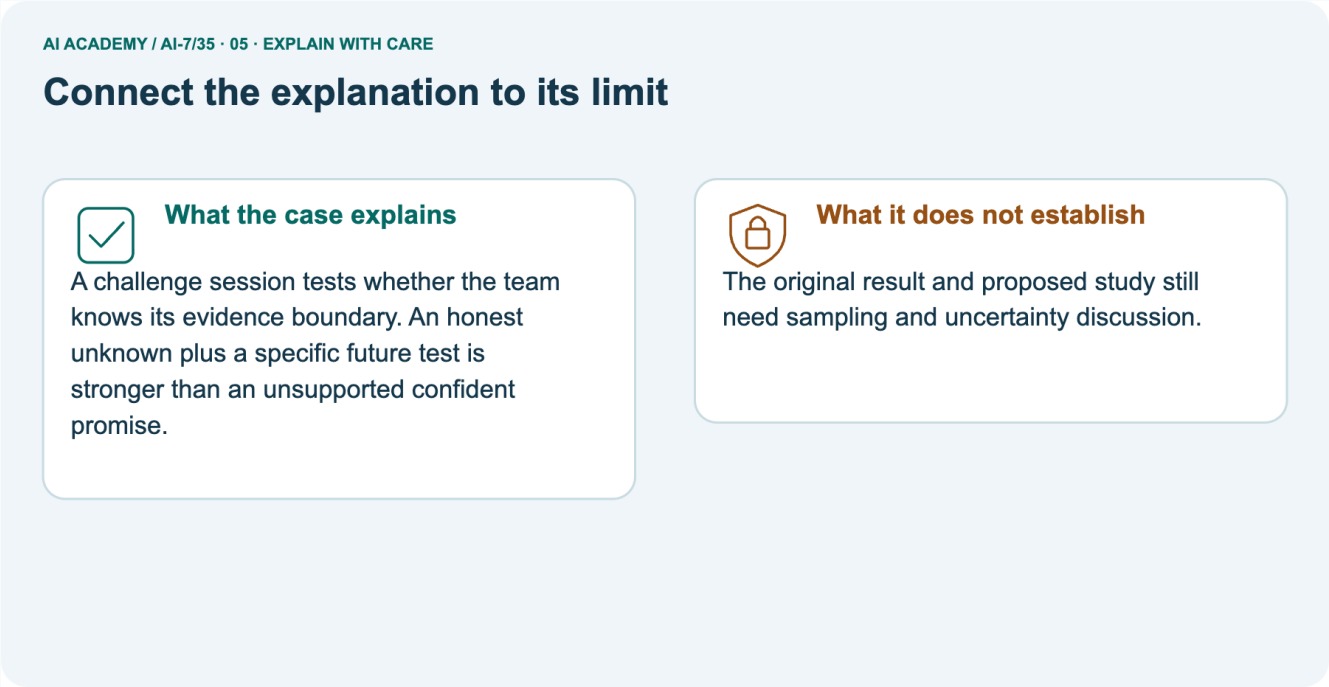
Expected result: That is one demonstration, not an estimate of broad camera-B reliability.
Reason: One selected **observation** does not represent the full proposed condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One selected **observation** does not represent the full proposed condition.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

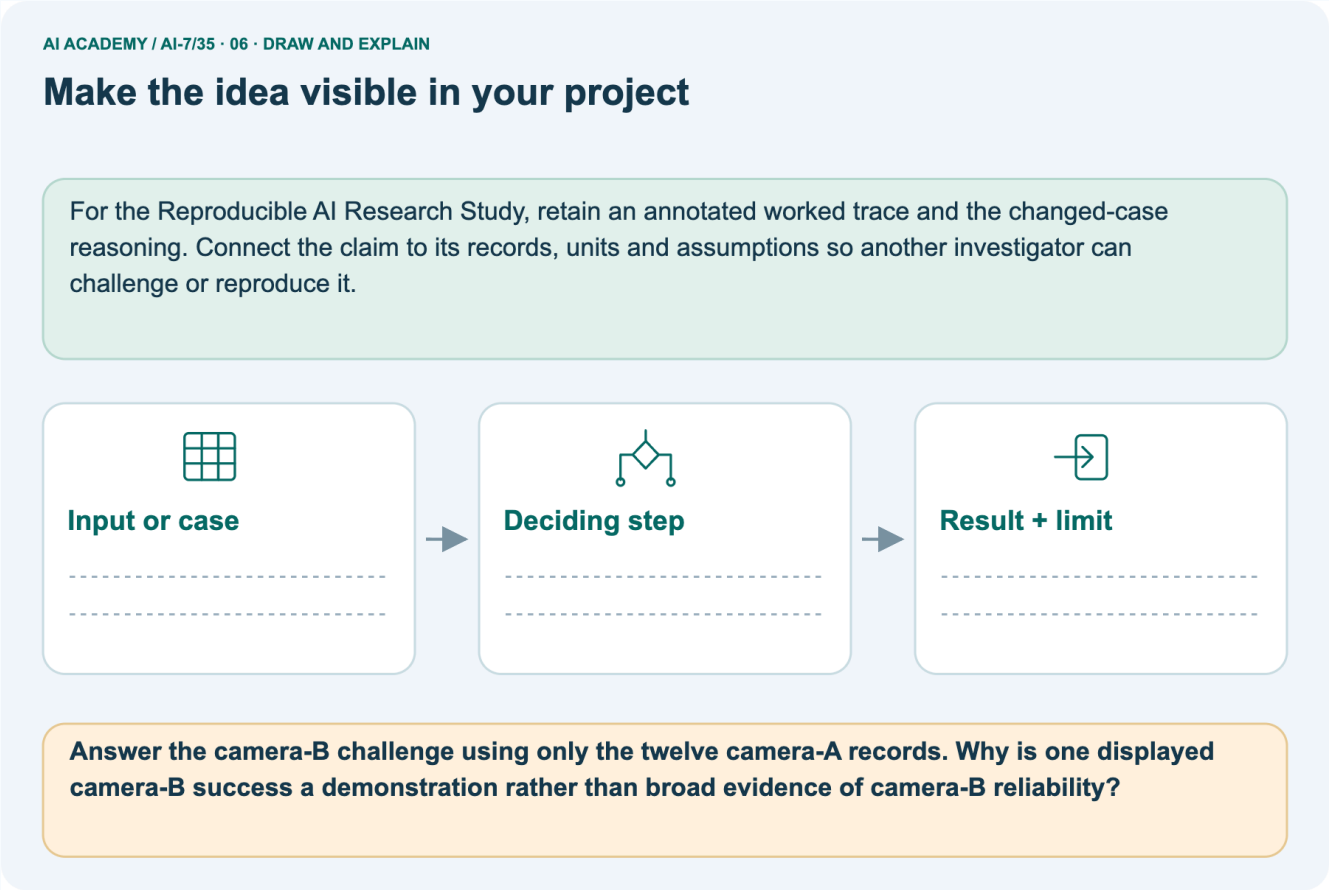
Limit: The original result and proposed study still need sampling and uncertainty discussion.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The original result and proposed study still need sampling and uncertainty discussion.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Report 10/12 correct on camera A. Say camera B performance is not measured. Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Changed case: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Presentation and challenge session: the case

02. Follow the deciding steps

03. Presentation and challenge session — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.